

Project Overview:

The QUELIFE Animal Shelter Organization operates animal sheltering operations in seven U.S. states: Wisconsin, Illinois, Missouri, Montana, California, Texas, and Nevada. The database is intended to automate and centralize tracking of animals, employees, donations, medical records, adoptions, and facility operations. Through the introduction of normalized relational tables and the explicit definition of functional dependencies (FDs), this database guarantees elimination of redundancy, data integrity, and effective querying.

The system employs Oracle Database for the management of structured data and handles different types of relationships (1:1, 1:M, M:N). All the relations are converted to BCNF or 3NF, depending on their dependencies and attributes.

Functional Dependencies (FDs) and Normalization

1. Facility

Attributes: Facility_ID (PK), Name, Location, PlayAreas, HealthCareUnits, OfficeSpace, TempControl

Functional Dependencies:

- Facility_ID → Name, Location, PlayAreas, HealthCareUnits, OfficeSpace, TempControl

Already in BCNF: All the non-key attributes are functionally dependent on the primary key.

2. Animal

Attributes: Animal_ID (PK), Name, Species, Breed, DOB, AcquisitionDate, AcquisitionMode, HealthStatus, Facility_ID (FK)

Functional Dependencies:

- Animal_ID → Name, Species, Breed, DOB, AcquisitionDate, AcquisitionMode, HealthStatus, Facility_ID
- (Species, Breed) → TypicalHealthRisks

In BCNF. (If TypicalHealthRisks were to be stored, it should be moved to a separate table to maintain BCNF.)

3. Adopter

Attributes: Adopter_ID (PK), Name, Address, Phone, Email, PreviousAdoptions, ApprovalStatus

Functional Dependencies:

- Adopter_ID → Name, Address, Phone, Email, PreviousAdoptions, ApprovalStatus

In BCNF

4. Adoption

Attributes: Adoption_ID (PK), Animal_ID (FK), Adopter_ID (FK), AdoptionDate, Fee, FollowUpStatus, Employee_ID (FK)

Functional Dependencies:

- Adoption_ID → Animal_ID, Adopter_ID, AdoptionDate, Fee, FollowUpStatus, Employee_ID
- Animal_ID → Adoption_ID (assuming one-to-one relationship)

In BCNF

5. Employee

Attributes: Employee_ID (PK), Name, Role, Facility_ID (FK), Education, Certification, CriminalRecord, Contact

Functional Dependencies:

- Employee_ID → Name, Role, Facility_ID, Education, Certification, CriminalRecord, Contact
- (Education, Certification) → Role

In BCNF

6. Sponsor

Attributes: Sponsor_ID (PK), Name, OrgName, ContactInfo, SponsorshipType

Functional Dependencies:

- Sponsor_ID → Name, OrgName, ContactInfo, SponsorshipType

In BCNF

7. Donation

Attributes: Donation_ID (PK), Sponsor_ID (FK), Facility_ID (FK), Amount, Date, Purpose

Functional Dependencies:

- Donation_ID → Sponsor_ID, Facility_ID, Amount, Date, Purpose

In BCNF

8. Medical_Record

Attributes: Record_ID (PK), Animal_ID (FK), CheckupDate, Diagnosis, Treatment, VetName, NextAppt

Functional Dependencies:

- Record_ID → Animal_ID, CheckupDate, Diagnosis, Treatment, VetName, NextAppt
- (Animal_ID, CheckupDate) → Diagnosis, Treatment, VetName, NextAppt

In BCNF

9. Supplier

Attributes: Supplier_ID (PK), Name, Contact, ItemType

Functional Dependencies:

- Supplier_ID → Name, Contact, ItemType

In BCNF

10. Facility_Supplier (Bridge Table for M:N)

Attributes: Facility_ID (FK), Supplier_ID (FK), SupplyDate

Functional Dependencies:

- (Facility_ID, Supplier_ID) → SupplyDate

In BCNF (Composite key)

Relationships after Normalization

- Facility–Animal: One-to-Many via Facility_ID
- Animal–Medical_Record: One-to-Many (one animal can have multiple records)
- Animal–Adoption: One-to-One (per adoption record)
- Adopter–Adoption: One-to-Many
- Employee–Adoption: One-to-Many (employee administers multiple adoptions)

- Facility–Employee: One-to-Many
- Sponsor–Donation: One-to-Many
- Facility–Donation: One-to-Many
- Facility–Supplier: Many-to-Many using Facility_Supplier
- Animal–Facility: Many-to-One

Advantages of Normalization

- Decreased Data Redundancy: Attributes separated into logically dependent tables.
- Update Anomalies removed: Every unit of data is stored only once.
- Improved Querying: Easier joins, correct reports, and improved performance.
- Better Integrity: Maintained through PKs, FKs, and BCNF/3NF compliance.

Updating Tables after Normalization

Facility Table (BCNF)

The screenshot shows the Live SQL interface with a SQL worksheet. The SQL code defines the Facility table with the following attributes and constraints:

```

1 CREATE TABLE Facility (
2   Facility_ID VARCHAR(50) PRIMARY KEY,
3   Facility_Name VARCHAR(100) NOT NULL,
4   Location VARCHAR(255) NOT NULL,
5   INT_of_Play_Areas INT CHECK (INT_of_Play_Areas >= 0),
6   Health_Care_Units INT CHECK (Health_Care_Units >= 0),
7   Office_Space_Available CHAR(1) CHECK (Office_Space_Available IN ('Y', 'N')),
8   Temperature_Control CHAR(1) CHECK (Temperature_Control IN ('Y', 'N'))
9 )
10
11

```

Below the code editor, the 'Script output' tab is active, showing the execution result: 'Table FACILITY created.' and 'Elapsed: 00:00:00.022'.

At the bottom of the interface, there are links for 'About Oracle', 'Contact Us', 'Legal Notices', 'Terms and Conditions', 'Your Privacy Rights', 'Delete Your Live SQL Account', and 'Cookie Preferences'. The copyright notice states 'Copyright © 2014, 2025 Oracle and/or its affiliates All rights reserved.' and the page number 'r29' is displayed in the bottom right corner.

Animal Table (BCNF)

The screenshot shows the Live SQL interface with a SQL worksheet. The SQL code defines the Animal table with the following attributes and constraints:

```
CREATE TABLE Animal (  
  Animal_ID VARCHAR(50) PRIMARY KEY,  
  Name VARCHAR(100) NOT NULL,  
  Species VARCHAR(50) NOT NULL,  
  Breed VARCHAR(50),  
  Date_of_Birth DATE,  
  Date_of_Acquisition DATE NOT NULL,  
  Mode_of_Acquisition VARCHAR(100) NOT NULL,  
  Health_Status VARCHAR(100) NOT NULL,  
  Facility_ID VARCHAR(50),  
  FOREIGN KEY (Facility_ID) REFERENCES Facility(Facility_ID) ON DELETE CASCADE  
);
```

The interface includes a top navigation bar with "Live SQL", "Worksheet", and "Library" tabs. Below the code editor, there are tabs for "Query result", "Script output", "DBMS output", "Explain Plan", and "SQL history". The "Script output" tab is active, displaying the message "Table ANIMAL created." and "Elapsed: 00:00:00.020". The footer contains links for "About Oracle", "Contact Us", "Legal Notices", "Terms and Conditions", "Your Privacy Rights", "Delete Your Live SQL Account", and "Cookie Preferences", along with the copyright notice "Copyright © 2014, 2025 Oracle and/or its affiliates All rights reserved." and the identifier "r29".

Employee Table (BCNF)

The screenshot shows the Live SQL interface with a SQL worksheet. The SQL code defines the Employee table with the following attributes and constraints:

```
CREATE TABLE Employee (  
  Employee_ID VARCHAR(50) PRIMARY KEY,  
  Name VARCHAR(100) NOT NULL,  
  Role VARCHAR(100) NOT NULL,  
  Facility_ID VARCHAR(50),  
  Education VARCHAR(100),  
  Certification VARCHAR(100),  
  Criminal_Record CHAR(1) CHECK (Criminal_Record IN ('Y', 'N')),  
  Contact_Info VARCHAR(255) UNIQUE NOT NULL,  
  FOREIGN KEY (Facility_ID) REFERENCES Facility(Facility_ID) ON DELETE SET NULL  
);
```

The interface includes a top navigation bar with "Live SQL", "Worksheet", and "Library" tabs. Below the code editor, there are tabs for "Query result", "Script output", "DBMS output", "Explain Plan", and "SQL history". The "Script output" tab is active, displaying the message "Table EMPLOYEE created." and "Elapsed: 00:00:00.022". The footer contains links for "About Oracle", "Contact Us", "Legal Notices", "Terms and Conditions", "Your Privacy Rights", "Delete Your Live SQL Account", and "Cookie Preferences", along with the copyright notice "Copyright © 2014, 2025 Oracle and/or its affiliates All rights reserved." and the identifier "r29".

Medical_Record Table (BCNF)

The screenshot shows the Oracle Live SQL interface. At the top, there's a navigation bar with "Live SQL", "Worksheet", and "Library" tabs. A "Return to Live SQL Classic" button is on the right, along with "Help and Feedback" and "Sign Out" links. Below the navigation bar is a toolbar with icons for running queries, saving, and other functions. The main area displays a SQL script for creating the "Medical_Record" table. The script is as follows:

```
37
38 CREATE TABLE Medical_Record (
39     Record_ID VARCHAR(50) PRIMARY KEY,
40     Animal_ID VARCHAR(50) NOT NULL,
41     Checkup_Date DATE NOT NULL,
42     Diagnosis VARCHAR(255),
43     Treatment VARCHAR(255),
44     Vet_Name VARCHAR(100) NOT NULL,
45     Next_Appointment DATE,
46     FOREIGN KEY (Animal_ID) REFERENCES Animal (Animal_ID) ON DELETE CASCADE
47 );
48
```

Below the script, there are tabs for "Query result", "Script output", "DBMS output", "Explain Plan", and "SQL history". The "Script output" tab is active, showing the message "Table MEDICAL_RECORD created." and "Elapsed: 00:00:00.030". At the bottom, there are links for "About Oracle", "Contact Us", "Legal Notices", "Terms and Conditions", "Your Privacy Rights", "Delete Your Live SQL Account", and "Cookie Preferences". The footer indicates "Copyright © 2014, 2025 Oracle and/or its affiliates All rights reserved." and a page number "r29".

Adopter Table (3NF)

The screenshot shows the Oracle Live SQL interface. At the top, there's a navigation bar with "Live SQL", "Worksheet", and "Library" tabs. A "Return to Live SQL Classic" button is on the right, along with "Help and Feedback" and "Sign Out" links. Below the navigation bar is a toolbar with icons for running queries, saving, and other functions. The main area displays a SQL script for creating the "Adopter" table. The script is as follows:

```
48
49
50 CREATE TABLE Adopter (
51     Adopter_ID VARCHAR(50) PRIMARY KEY,
52     Name VARCHAR(100) NOT NULL,
53     Address VARCHAR(255) NOT NULL,
54     Phone VARCHAR(20) NOT NULL,
55     Email VARCHAR(100) UNIQUE NOT NULL,
56     Previous_Adoptions INT CHECK (Previous_Adoptions >= 0),
57     Approval_Status VARCHAR(50) CHECK (Approval_Status IN ('Approved', 'Pending', 'Rejected'))
58 );
59
60
```

Below the script, there are tabs for "Query result", "Script output", "DBMS output", "Explain Plan", and "SQL history". The "Script output" tab is active, showing the message "Table ADOPTER created." and "Elapsed: 00:00:00.026". At the bottom, there are links for "About Oracle", "Contact Us", "Legal Notices", "Terms and Conditions", "Your Privacy Rights", "Delete Your Live SQL Account", and "Cookie Preferences". The footer indicates "Copyright © 2014, 2025 Oracle and/or its affiliates All rights reserved." and a page number "r29".

Adoption Table (BCNF)

Live SQL

Worksheet

Library

Return to Live SQL Classic

Help and Feedback

Sign Out

[SQL Worksheet]*

```
61 CREATE TABLE Adoption (
62     Adoption_ID VARCHAR(50) PRIMARY KEY,
63     Animal_ID VARCHAR(50) NOT NULL,
64     Adopter_ID VARCHAR(50) NOT NULL,
65     Adoption_Date DATE NOT NULL,
66     Adoption_Fee DECIMAL(10,2) CHECK (Adoption_Fee >= 0),
67     Follow_up_Status VARCHAR(100),
68     Employee_ID VARCHAR(50),
69     FOREIGN KEY (Animal_ID) REFERENCES Animal(Animal_ID) ON DELETE CASCADE,
70     FOREIGN KEY (Adopter_ID) REFERENCES Adopter(Adopter_ID) ON DELETE CASCADE,
71     FOREIGN KEY (Employee_ID) REFERENCES Employee(Employee_ID)
72 );
```

Query result

Script output

DBMS output

Explain Plan

SQL history

Table ADOPTION created.

Elapsed: 00:00:00.034

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Sponsor Table (BCNF)

Live SQL

Worksheet

Library

Return to Live SQL Classic

Help and Feedback

Sign Out

[SQL Worksheet]*

```
74
75
76 CREATE TABLE Sponsor (
77     Sponsor_ID VARCHAR(50) PRIMARY KEY,
78     Name VARCHAR(100) NOT NULL,
79     Organization_Name VARCHAR(100),
80     Contact_Info VARCHAR(255) NOT NULL,
81     Donation_Amount DECIMAL(10,2) CHECK (Donation_Amount >= 0),
82     Sponsorship_Type VARCHAR(100) NOT NULL
83 );
84
85
```

Query result

Script output

DBMS output

Explain Plan

SQL history

Table SPONSOR created.

Elapsed: 00:00:00.023

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Donation Table (BCNF)

Live SQL

Worksheet

Library

Return to Live SQL Classic

Help and Feedback

Sign Out

[SQL Worksheet]*

```
87
88 CREATE TABLE Donation (
89     ... Donation_ID VARCHAR(50) PRIMARY KEY,
90     ... Sponsor_ID VARCHAR(50) NOT NULL,
91     ... Facility_ID VARCHAR(50) NOT NULL,
92     ... Amount DECIMAL(10,2) CHECK (Amount >= 0),
93     ... Donation_Date DATE NOT NULL,
94     ... Purpose VARCHAR(255),
95     ... FOREIGN KEY (Sponsor_ID) REFERENCES Sponsor(Sponsor_ID) ON DELETE CASCADE,
96     ... FOREIGN KEY (Facility_ID) REFERENCES Facility(Facility_ID) ON DELETE CASCADE
97 ) ;
98
```

Query result

Script output

DBMS output

Explain Plan

SQL history

Table DONATION created.

Elapsed: 00:00:00.031

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Supplier Table (BCNF)

Live SQL

Worksheet

Library

Return to Live SQL Classic

Help and Feedback

Sign Out

[SQL Worksheet]*

```
100
101 CREATE TABLE Supplier (
102     ... Supplier_ID VARCHAR(50) PRIMARY KEY,
103     ... Name VARCHAR(100) NOT NULL,
104     ... Contact_Info VARCHAR(255) NOT NULL,
105     ... Supply_Type VARCHAR(100) NOT NULL
106 ) ;
107
```

Query result

Script output

DBMS output

Explain Plan

SQL history

Table SUPPLIER created.

Elapsed: 00:00:00.022

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My Contributions:

- Designed the ERD according to the given project description.
- Defined all the entity sets and relationships, i.e., attributes, cardinalities, and participation.
- Translated the ERD to a relational schema and normalized to BCNF.
- Written and ran DDL scripts with constraints (PK, FK, NOT NULL, UNIQUE, CHECK).
- Wrote general-purpose CHECK constraints to impose business rules.
- Included screenshots of schema creation and validation in Oracle SQL Developer.